Advancements in LiDAR technology for precision water resource management

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Abstract

Progressions in Light Detection and Ranging (LiDAR) technology have ushered in a new era of precision water resource management, offering an unprecedented level of detail and accuracy. This review explores the transformative role of LiDAR in hydrological studies, highlighting its capacity to generate high-resolution topographical maps and three-dimensional (3D) models of water bodies and surrounding

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Advances in Chemical Pollution, Environmental Management and Protection
ISSN 2468-9289, https://doi.org/10.1016/bs.apmp.2025.11.013

landscapes. Through a series of case studies and practical applications, the narrative demonstrates how LiDAR data facilitates the meticulous monitoring of watershed dynamics, floodplain mapping, and the assessment of aquatic habitats. Besides, the technical nuances of LiDAR, from data acquisition through airborne and terrestrial platforms to sophisticated data processing techniques, was delved into. It emphasizes the significance of LiDAR-derived digital elevation models (DEMs) in simulating hydrological processes and enhancing the predictive accuracy of flood models. Over and above that, the integration of LiDAR with other remote sensing technologies and Geographic Information Systems (GIS) is examined, illustrating a holistic approach to water resource management. Challenges associated with LiDAR deployment, such as cost, data volume, and the necessity for specialized skills, are candidly discussed. Innovations aimed at overcoming these obstacles are also presented, showcasing the continuous evolution of LiDAR technology in the face of climate change.

Keywords: LiDAR, Water resource management, Floodplain mapping, Hydrological modeling, Geographic information systems.

1. Introduction

Light Detection and Ranging (LiDAR) systems are mainly manufactured in technologically advanced regions such as the United States, Europe (notably Germany and France), China, and Japan by specialized geospatial and sensor companies. These regions dominate due to strong electronics industries, advanced optics manufacturing, and sustained investment in aerospace, autonomous systems, and remote sensing technologies. Improvements in remote sensing technologies have greatly remodeled environmental monitoring and management practices.[1,2,3] LiDAR technology, in particular, has surfaced as a decisive tool in precision water resource management,[4] as it offers high-resolution spatial data that enhances the understanding of water systems.[5] Previous research has highlighted the effectiveness of LiDAR in mapping floodplains,[6] river morphology assessment, and the monitoring of wetland ecosystems.[7] Studies have demonstrated its capability to provide detailed elevation models, which are essential for flood risk assessments and habitat analysis. Recent developments later pushed the boundaries of LiDAR applications, by integrating it with other remote sensing technologies and advanced analytical tools. To give an instance, integration with satellite imagery and Geographic Information Systems (GIS) now allowed for more comprehensive water resource mapping and management.[8,9] Although, existing research has established the foundational applications of LiDAR in environmental monitoring, there remains a need to explore its full potential in assessing water quality, climate change impacts adaptation, and the optimization of water resource management strategies.[10] It is already being used in South Africa for flood and erosion assessment using drones and airborne systems, and in aerial survey projects like terrain mapping along the Niger River in West Africa, which supports catchment and river studies. Emerging applications also include integration with digital twins (DT) for river basins such as the Limpopo, where high-resolution LiDAR and multispectral data help model surface features for planning and decision-making. Other tools include satellite optical remote sensing (e.g., Landsat, Sentinel), radar systems e.g., Synthetic Aperture Radar (SAR), and ground-based Global Positioning System (GPS)/total station surveys, which are useful for broad monitoring but often lack fine elevation detail or are affected by cloud cover. LiDAR is superior because it provides very high-resolution, accurate 3D elevation data, can penetrate vegetation canopies, and enables precise floodplain, drainage, and water-infrastructure analysis. Hence, the objectives of this study include exploring advancements in LiDAR technology, evaluating its impact on water quality assessments, and assessing its role in climate change adaptation for water resources. Moreover, integrating LiDAR with emerging technologies and addressing challenges associated with its application will be examined. The current examination aims at providing a comprehensive understanding of how LiDAR can enhance precision in water resource management and guide future research and practical applications in this field.

2. Technical fundamentals of LiDAR

LiDAR is a remote sensing method that utilizes laser light to measure variable distances to the earth. It operates by emitting laser pulses from a sensor towards the ground and records the time it takes for the light to bounce back after hitting an object or surface. Fundamental components of a LiDAR system include (Fig. 1): a laser,[11] a scanner,[10] and a specialized GPS receiver.[12] When laser pulses are emitted, the scanner directs the laser across the target area, and the GPS receiver accurately records the position and movement of the LiDAR sensor. Operation of LiDAR relies on precise timing and positioning. Each laser pulse travels at the speed of light, and the time delay between emission and return is measured to calculate the distance traveled. It is known as time-of-flight measurement, because the process enables the creation of detailed 3D models of the scanned

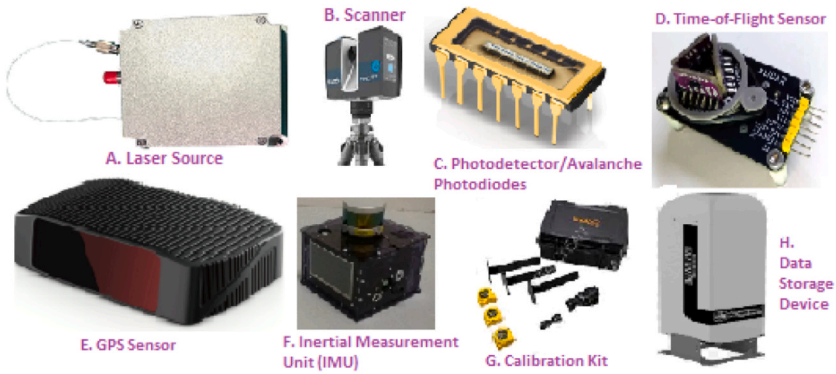

Fig. 1 Components of LiDAR systems.

environment.[13,14] Modern LiDAR systems usually emit hundreds of thousands of pulses per second, so as to boost data density and resolution.

A critical aspect of LiDAR technology is its ability to penetrate vegetation and provide high-resolution topographic data even in densely forested areas.[15–17] Achieving this capability involves multiple returns, where the sensor records the reflected signals from different surfaces, such as the top of the canopy[18,19] and the ground. It is particularly useful in water resource management, as understanding the terrain beneath vegetation is crucial for accurate modeling and analysis.[20] Data collected by LiDAR sensors are processed to generate various types of output, including DEMs,[17,21] digital surface models (DSMs),[22] and point clouds.[23] DEMs represent the bare ground surface without any objects,[24,25] whereas DSMs include all the objects on the surface, such as buildings and vegetation.[26] Point clouds are collections of points that represent the 3D positions of objects in the scanned area, which can be used for further analysis and visualization.[14,15] LiDAR systems can be deployed from aircraft, drones,[27] and ground-based vehicular platforms,[15,25] each offering different advantages depending on the application.

3. Applications in water resource management

LiDAR technology is an influential tool in water resource management,[22]due to its ability to offer high-resolution, accurate topographic and bathymetric data,[28,29] Floodplain mapping and flood risk assessment are primary applications. Creating detailed DEMs with LiDAR will help identify flood-prone areas according to Kilpatrick,[24] further enhancing planning and mitigation strategies. High-resolution data also supports the development of hydraulic models that predict flood behavior under various scenarios. Watershed monitoring and management benefit immensely from LiDAR's precision.[18] Accurate mapping of watershed boundaries, drainage patterns, and stream networks is essential for effective water resource planning.[30] LiDAR data identify potential erosion zones and sediment transport paths, which are critical for maintaining water quality and ecosystem health. Over and above that, LiDAR-derived models assist in designing and implementing best management practices to reduce erosion and sedimentation.

River and stream management applications leverage LiDAR for detailed information on channel morphology and riparian vegetation.[31] It is mentioned in Buytaert et al.[32] that understanding channel dynamics and vegetation patterns is momentous for maintaining healthy aquatic habitats and ensuring sustainable water resources. LiDAR data facilitate the assessment of streambank stability and the identification of areas requiring restoration or protection, in order to support efforts made atenhancing habitat quality, improving water quality, and maintaining biodiversity. Reservoir and dam management also reap significant benefits from LiDAR technology.[33] Accurate topographic data are essential during the design and maintenance of these structures. LiDAR surveys provide detailed information on reservoir capacity, sedimentation rates, and potential hazards, aiding in optimizing reservoir operations, ensuring structural safety, and extending the lifespan of critical infrastructure.[34]

Coastal[35] and estuarine management applications use LiDAR to monitor highly dynamic areas susceptible to erosion, sea-level rise, and storm surges.[36] High-resolution coastal maps generated from LiDAR data monitors shoreline changes, assess coastal erosion, and plan for climate change impacts. In order to develop coastal protection measures and manage coastal ecosystems effectively, those maps are essential.[37] Irrigation management optimization is another application of LiDAR-derived data.[38] Accurate elevation models assist in designing efficient irrigation systems by identifying optimal water distribution paths and minimizing water loss.[39] Precision agriculture,[9,40,41] benefits from LiDAR by ensuring that water resources are used efficiently, which is vital in arid and semi-arid regions where water scarcity poses real concern, as it also affects crops.[42,43] Groundwater management also sees benefits from LiDAR.[44] Simply because, it provide exhaustive surface elevation data helps in digesting the relationship between surface water[45,46] and groundwater. Though, it is essential for recharge studies, aquifer mapping, and sustainable groundwater extraction practices, accurate topographic data will assist during the identification of potential recharge areas and the effective management of groundwater resources.

3.1 Impact of LiDAR on water quality assessment

LiDAR technology chiefly augment water quality assessment (WQA) by providing detailed and accurate data that traditional methods often miss. One of its key contributions is in mapping and monitoring watershed areas.[33] Accurate topographic data from LiDAR seldom identify runoff from agricultural lands,[47] industrial sites, urban areas, as well as other sources of pollution.[48] Knowing precisely where pollutants originate gives room for targeted interventions to prevent contaminants from reaching water bodies. Another critical area where LiDAR impacts water quality is in the assessment of erosion and sedimentation patterns. Sediment is a major pollutant in many water bodies, as it affects aquatic habitats and water quality. LiDAR provides high-resolution elevation data[49] that helps identify erosion hotspots and sediment transport pathways. This information is vital for the development of effective soil conservation practices and sediment management strategies,[50] and the reduction in the amount of sediment that enters rivers, lakes, and reservoirs.[51] Riparian zone management benefits greatly from LiDAR technology. Riparian zones, the interface between land and a river or stream, play an observable role in maintaining water quality. They act as buffers that filter out pollutants before they reach the water. LiDAR data helps map these zones with high precision, so that better management practices to preserve or restore these natural filters are enabled.[52] As it is, properly managed riparian zones reduce nutrient loads and improve overall water quality.

LiDAR also aids in the detection and monitoring of algal blooms, which are prime concerns for water quality. Excess nutrients, particularly nitrogen and phosphorus, often lead to harmful algal blooms that deplete oxygen in water and produce toxins. High-resolution LiDAR data then identify areas contributing to nutrient loading, such as agricultural fields with heavy fertilizer use[53] or wastewater discharge sites. With this info, measures can be implemented to control nutrient inputs, thereby mitigating algal blooms. Wetland mapping and monitoring is another area were LiDAR impacts WQA. Wetlands are natural water purifiers capable of trapping sediments and absorbing pollutants. Accurate elevation data from

LiDAR most times, enables the precise delineation of wetlands, monitors changes over time, and assess the health of these crucial ecosystems.[19] Protecting and restoring wetlands based on LiDAR data contributes to improved water quality, due to the fact that it enhances their natural filtration functions.

Urban water quality management also benefits from LiDAR technology.[34] Urban areas contribute voluminous amounts of pollutants to water bodies through stormwater runoff. LiDAR data then supports the design and implementation of green infrastructure, such as bioswales and rain gardens, which mitigate the impact of stormwater runoff. Detailed topographic information is another way to ensure whether these systems are optimally placed and effective in filtering pollutants before they reach water bodies.[28] In agricultural settings, LiDAR's role is in precision farming practices that benefit water quality.[38,40] The provision of detailed terrain models enables LiDAR to aid in the design of efficient irrigation systems[42] that minimize water and fertilizer use. To improve water quality, reducing excess irrigation and fertilizer runoff directly decreases the amount of pollutants entering water bodies. LiDAR's role in floodplain mapping also indirectly benefits water quality. Accurate floodplain maps help manage land use in flood-prone areas, and prevent the development that could increase pollution risks during flooding events. Managing floodplains effectively is said to reduce the likelihood of pollutants being washed into water bodies during floods.

3.2 Role of LiDAR in climate change adaptation for water resources

Adapting water resources to climate change requires innovative technologies, and LiDAR is heavily involve in this effort. High-resolution topographic data from LiDAR systems may identify vulnerable areas prone to flooding, for better planning and mitigation. Accurate elevation models still aid in predicting flood zones,[54,55] in order to facilitate the ability to protect communities and infrastructure. Increased precipitation and extreme weather events, which are hallmarks of climate change, necessitate robust watershed management. LiDAR provides essential data for mapping watersheds, analyzing drainage patterns, and modeling water flow.[24,54] Such information is vital when designing interventions to manage runoff, reduce erosion, and prevent sedimentation in water bodies. Coastal areas facing rising sea levels and increased storm intensity, benefit sizeably from LiDAR. Coastal erosion, storm surge, and sea-level rise can be monitored and modeled accurately, informing strategies for shoreline protection and habitat restoration. Coastal managers may decide to use this data to develop effective responses to mitigate the impacts of climate change on coastal water resources. Agricultural practices must also adapt to changing climate conditions, and precision farming supported by LiDAR data is key.[56] Terrain models guide efficient irrigation system design and optimizes water use and waste reduction. In areas facing increased drought risk,[57,58] it helps conserve water resources and ensures sustainable agricultural practices.[39,47]

Groundwater resources, often impacted by climate variability, can be better managed using LiDAR. Surface elevation data aids in identifying recharge areas[52] and understanding the interaction between surface water and groundwater. According to Ibrahim et al.,[59] this is central for sustainable groundwater management, so that aquifers are replenished and not over-extracted. Urban areas, with their unique challenges, also see significant benefits. Stormwater management systems, essential for coping with increased rainfall and runoff emphasized in Doost et al.,[57] are sometimes optimized using precise topographic data. Green infrastructure, such as permeable pavements and green roofs, can be strategically placed to enhance stormwater absorption and reduce urban flooding. Wetlands, acting as natural buffers, are critical in climate adaptation. Accurate mapping and monitoring of wetlands ensure their preservation and restoration.[6] Healthy wetlands absorb floodwaters, filter pollutants, and provide habitat for wildlife, all crucial functions in a changing climate.

4. Integration with other technologies

Coalescing LiDAR with other advanced technologies naturally foster its effectiveness in precision water resource management.[60] Again, combining LiDAR data with GIS trigger some sophisticated spatial analyses and visualizations. GIS platforms can overlay LiDAR-derived elevation models with other spatial data, such as land use and vegetation cover, to provide a comprehensive understanding of water resources.[45,56,61] Remote sensing technologies complement LiDAR by offering broader coverage and different data types.[1] Satellites[62,63] and aerial[64] imagery provide large-scale views of water bodies, vegetation health, and land surface changes. When paired with LiDAR's high-resolution data, the aforementioned technologies create a robust framework for monitoring and managing water resources effectively. Unmanned Aerial Vehicles (UAVs) highlighted by

Ballabh et al.[40] and Mishra & Rai,[64] are commonly referred to as drones, and are increasingly being used to deploy LiDAR systems in hard-to-reach areas. Drones offer flexibility and cost-effectiveness, as well as frequent and targeted data collection.[65,66] The combination gives real-time monitoring and rapid response to changing environmental conditions. Sharma et al.[34] and Wada et al.[67] found that the integration with hydrological models undeniably supports water resource management. LiDAR data feeds into various models to simulate water flow, predict flooding events, and manage water distribution. In that case, accurate topographic information ensures that models are precise, for decision-making and planning.

Cutting-edge data processing and machine learning (ML) algorithms[68] further expand the capabilities of LiDAR. ML techniques analyze vast amounts of LiDAR data to identify patterns and predict trends.[69] This enables more proactive and efficient management of water resources, to address issues before they escalate. Incorporating Internet of Things (IoT) devices creates a network of sensors that continuously collect environmental data.[70] IoT sensors monitor water quality, temperature, and flow rates parameters.[71] When integrated with LiDAR data, this information provides a real-time, dynamic picture of water resource conditions, further simplifying the ability to manage them adaptively. Collaboration with cloud computing platforms[72] facilitates the storage, processing, and sharing of large LiDAR datasets. Cloud-based systems enable seamless access to data for stakeholders,[16] to promote collaborative water resource management. Real-time data processing in the cloud also ensures timely analysis and dissemination of information. Solar-powered and autonomous systems extend the reach of LiDAR in remote and off-grid locations. Several or similar systems can operate independently, collecting data over long periods without human intervention. It ensures continuous monitoring of water resources, even in areas that are difficult to access,

5. Challenges and solutions

Several setbacks may be encountered in this respect, as shown in Table 1. However, a well-maintained equipment operates more reliably to minimize downtime and associated costs.[80] Combining LiDAR with other remote sensing methods, such as aerial imagery,[43] further improve accuracy in challenging environments.

Table 1 Some identified demerits of LiDAR technologies and their solutions.

No.	challenges	Solutions	Literature source
1.	High Initial Costs	Utilizing government grants, partnerships, and phased implementation to spread out expenses.	LIDAR[73]
2.	Data Processing Complexity	Employing advanced software, machine learning, and cloud computing for efficient data handling.	Kanga[16]
3.	Limited Expertise	Providing training programs, workshops, and collaborating with academic institutions to build expertise.	Hopkinson et al.,[74,75]
4.	Weather Dependency	Scheduling surveys during optimal weather conditions and using drones for flexibility.	Debell et al.,[21,75]
5.	Data Integration	Developing standardized protocols and using interoperable systems to ensure seamless data integration.	Dowman[76]
6.	Equipment Maintenance	Implementing regular maintenance schedules and investing in durable, high-quality equipment.	SICK[77]
7.	Accuracy in Dense Vegetation	Using multi-return LiDAR technology and combining with other remote sensing methods for improved accuracy.	Sharma et al.,[35]
8.	Legal and Regulatory Issues	Ensuring compliance with local regulations through thorough research and legal consultation.	Crane et al.,[78,79]

Consistent data formats and standards facilitate the combination of different types of data, which enhances the overall analysis and decision-making processes. Furthermore, according to Maurya et al.,[81] drones provide additional flexibility for data collection during short windows of favorable weather, thus reducing delays and improving data reliability. It is foretold that creating a well-trained workforce ensures that LiDAR technology is used effectively and to its full potential in managing water resources. As mentioned in Table 1, spreading out costs over time and leveraging external funding sources make the technology more accessible to various water management organizations.

6. Next-generation developments and technological revolutions

LiDAR know-how has seriously transformed remote sensing by delivering high-resolution,[49] precise spatial data crucial for water resource management. Future advancements in LiDAR technology promise to further multiply its utility in this field. One key area of innovation is the improvement of sensor capabilities, with developments in laser and detector technologies expected to provide even finer spatial resolution[82] and greater accuracy. This progress will enhance the precision of mapping river networks, wetlands, and floodplains, as mentioned by Kumar et al.,[58] and later leads to more accurate modeling and management of water resources.[59] On top of that, integrating LiDAR with multi-spectral and hyperspectral imaging will offer deeper insights into vegetation health and water quality,[4,83] also enabling more comprehensive assessments of aquatic ecosystems.[84]

The future of LiDAR also lies in its integration with other technologies. Fusing LiDAR data with satellite imagery and aerial photography will create more accurate land and water resource maps, while advanced GIS will facilitate the creation of detailed 3D models of watersheds[61] and floodplains. The two models will support better decision-making in water management, including flood risk assessment and infrastructure planning. Moreover, innovations in real-time data acquisition and processing will allow for continuous monitoring of water bodies, and provide up-to-date information on water levels,[62] flow rates, and landscape changes. Cloud computing platforms will further enhance this capability by enabling efficient processing and accessibility of large volumes of LiDAR data, where timely and informed decision-making is facilitated.

Developments in UAV[31,46] and mobile LiDAR systems will expand data collection possibilities in diverse and challenging environments. UAVs equipped with LiDAR sensors will be able to access remote or hazardous areas (e.g., landslide location) that traditional methods might miss,[13,20,85,] while mobile LiDAR systems mounted on vehicles or boats will capture detailed data along river corridors and other linear features. Monitoring extensive regions and evaluating the effects of infrastructure on water resources are key advantages of these systems.[34,86] At length, enhanced data analytics and modeling techniques will be pivotal in extracting meaningful insights from LiDAR data. ML and artificial intelligence (AI) will automate feature identification, land cover classification, and trend prediction, for improved flood risk assessments and water quality forecasts.[65,69,87] Advanced modeling tools will also enable the simulation of various climate change impacts and land use changes scenarios, in order to support proactive management strategies and better preparedness for future challenges. Sit et al.[88] and Dixon & Uddameri[8] critically reviewed the role of artificial neural network (ANN) modelling and fuzzy set theory, respectively, in water resources and hydrology.

6.1 Potential applications of LiDAR in climate-smart water management

LiDAR technology has great potential in climate-smart water management.[89] First, it provides high-resolution topographic data that help in mapping watersheds, monitoring water bodies, and assessing flood risks. Using LiDAR in view of researchers, is an opportunity to be able to analyze river flow patterns and identify areas prone to erosion for better planning for flood control and water conservation. Secondly, LiDAR supports groundwater management.[90] It helps detect changes in land elevation, which can indicate groundwater depletion. Identification of suitable locations for water storage and recharge projects is assisted by this technology. In agriculture, LiDAR improves irrigation planning by mapping soil moisture levels and terrain slopes.[91,92] As a result, it allows farmers to use water efficiently, for the purpose of reducing waste and enhancing crop yield. LiDAR is valuable in monitoring the effects of climate change on water resources. It helps track changes in glacier volume, river paths, and coastal erosion. Related data obtained supports climate adaptation strategies and provide accurate predictions of water availability.

7. Case studies

LiDAR technology has been instrumental in enhancing water resource management through detailed spatial data and improved modeling techniques. Table 2 display some of the illustrative case studies that demonstrate the application and benefits of advanced LiDAR systems in this field.

Table 2 Case studies on advanced LiDAR technology for precision water resource management.

Elements	**Case study**				
	A	**B**	**C**	**D**	**E**
Title	Flood Risk Assessment in the Mississippi River Basin	Monitoring Wetland Changes in the Everglades National Park	Riverbank Erosion Control in the Rhine River	Urban Water Management in Singapore	Watershed Management in the Amazon Basin
Objective	Assess flood risk and create accurate floodplain maps.	Track changes in wetland areas and vegetation over time.	Evaluate and control erosion along riverbanks to protect infrastructure.	Optimize water management and stormwater runoff in a rapidly urbanizing city.	Manage and conserve watershed areas to prevent deforestation and maintain biodiversity.
Technology Used	High-resolution airborne LiDAR and satellite imagery integration.	UAV-based LiDAR and multi-spectral imaging.	Mobile LiDAR systems mounted on boats and aerial platforms.	Ground-based and UAV-based LiDAR combined with real-time monitoring systems.	Airborne LiDAR and satellite remote sensing combined with GIS.
Data Collected	Elevation data, floodplain delineation, vegetation mapping, and hydrological modeling.	Detailed elevation changes, vegetation health, water levels, and land cover classification.	Riverbank topography, erosion rates, sediment transport, and vegetation mapping.	Urban drainage patterns, water quality, flood risk areas, and infrastructure impacts.	Watershed boundaries, forest cover changes, soil erosion rates, and river flow patterns.

(continued)

Table 2 Case studies on advanced LiDAR technology for precision water resource management. (*cont'd*)

Elements	Case study				
	A	B	C	D	E
Key Findings	Identified high-risk flood zones and improved floodplain management strategies.	Detected significant changes in wetland extent and vegetation health, informing conservation efforts.	Successfully mitigated erosion and protected critical infrastructure through targeted interventions.	Enhanced stormwater management practices and reduced urban flooding incidents.	Provided insights for better watershed conservation practices and reduced deforestation.
Impact	Improved flood preparedness and emergency response, reducing flood damage.	Enabled more effective wetland preservation and management strategies.	Strengthened riverbank stability and safeguarded valuable infrastructure.	Improved urban water management and resilience to extreme weather events.	Advanced conservation measures and better management of natural resources.
Future Directions	Expand to include more dynamic hydrological models and integrate with real-time data sources.	Utilize higher resolution LiDAR and further integrate with habitat modeling tools.	Implement additional erosion control measures based on detailed LiDAR data.	Develop more sophisticated models incorporating climate change projections and urban growth.	Increase collaboration with local communities and integrate indigenous knowledge in management strategies.
Source	Rango, Anderson[93]	Hong, Wdowinski[94,95]	Krapesch et al.,[96]	Banerjee et al.,[97,98]	Braga et al.,[99,100]

Evidently, the case studies in Table 2 illustrate the diverse applications of advanced LiDAR technology in water resource management. It also highlight its role in improving flood risk assessments, monitoring environmental changes, and enhancing urban[4] and watershed management.[5] Continuous evolution of LiDAR technology and its integration with other tools promise further advancements in these areas.

8. Conclusion

In summary, advancements in LiDAR technology have profoundly influenced the field of water resource management, offering unprecedented levels of detail and accuracy in spatial data collection. Through the exploration of its technical fundamentals, diverse applications, and integration with other technologies, it becomes clear that LiDAR is pivotal in enhancing water quality assessments and adapting to climate change impacts. The ability to generate high-resolution elevation models and monitor dynamic changes in water systems has improved flood risk management, informed conservation efforts, and optimized urban water management practices. While challenges such as data processing and integration remain, ongoing innovations promise to address these issues and expand LiDAR's capabilities. Future research should continue to focus on refining LiDAR technologies and exploring new applications to address emerging water management challenges. As the field progresses, the integration of LiDAR with other advanced tools and methodologies will undoubtedly lead to more effective and sustainable management of our precious water resources.

Declaration of generative AI and AI-assisted technologies in the writing process

This work has been prepared with the assistance of generative AI in the writing process, specifically for drafting and language refinement. However, the final content has been extensively reviewed, edited, and verified by the authors to ensure accuracy, originality, and compliance with academic standards.

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